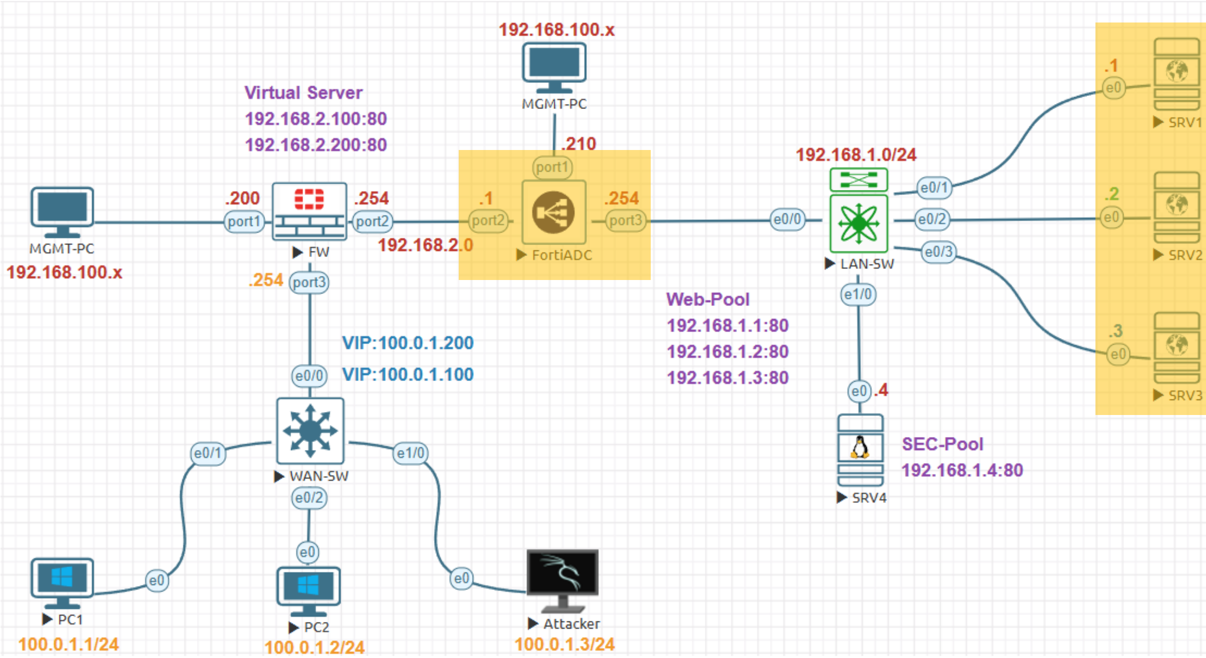
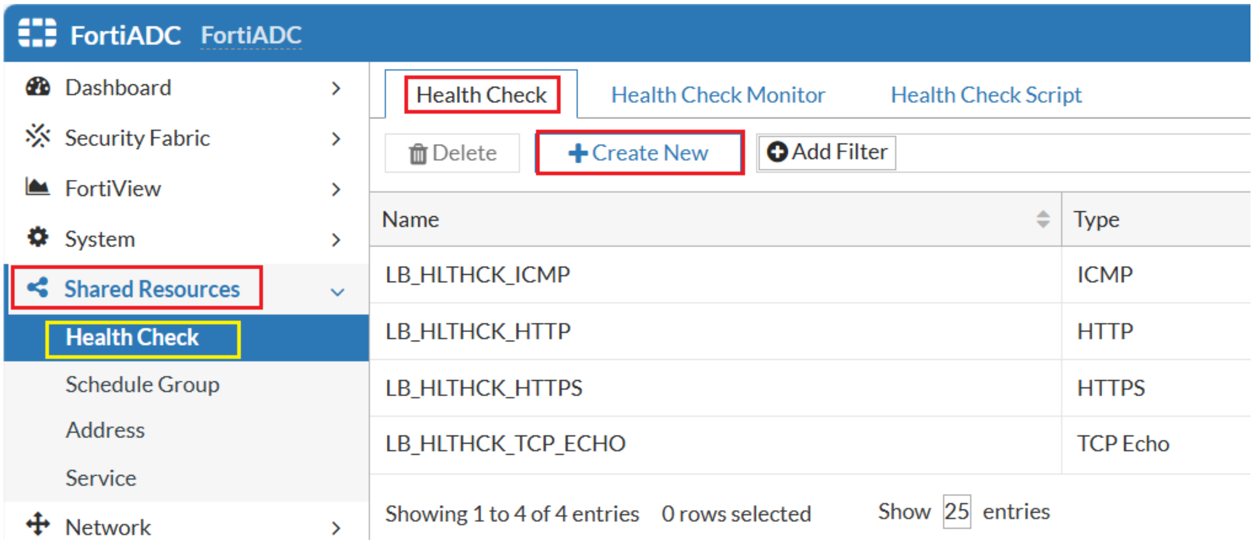
ICMP Health Check Lab:

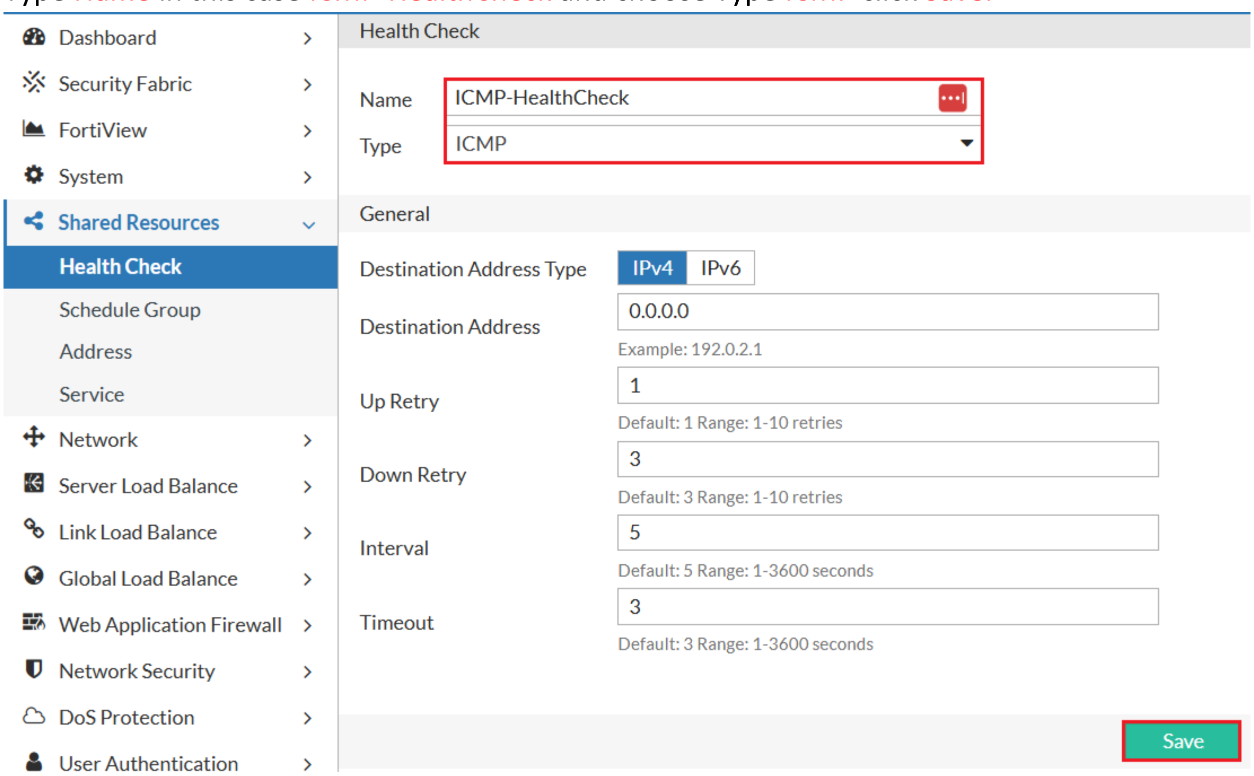


Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

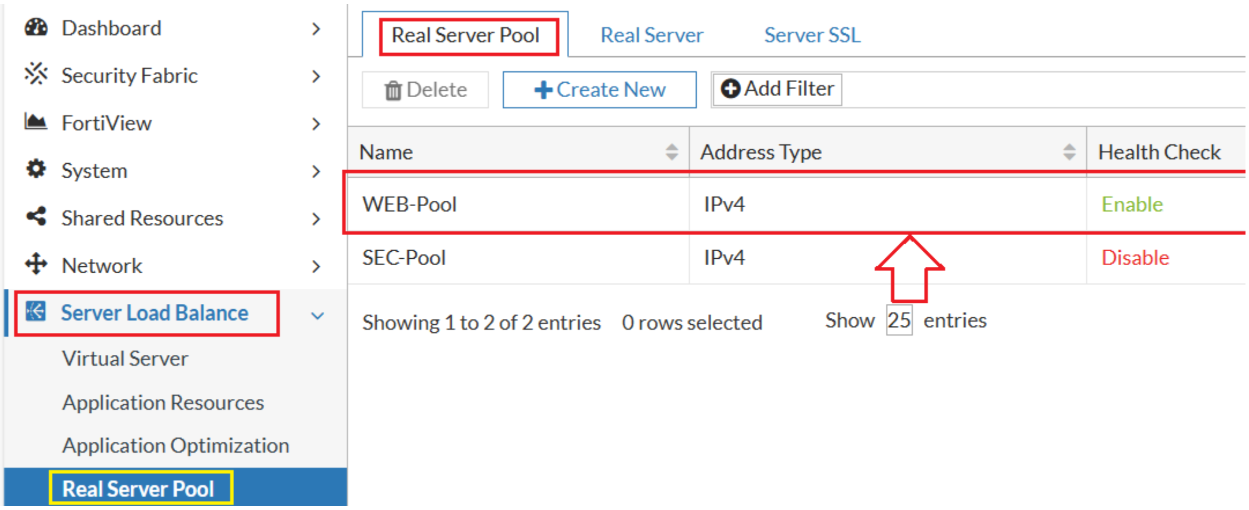
Let's create our own custom Health Check Navigate to Shared Resources>Health Check>Health Check click on Create New



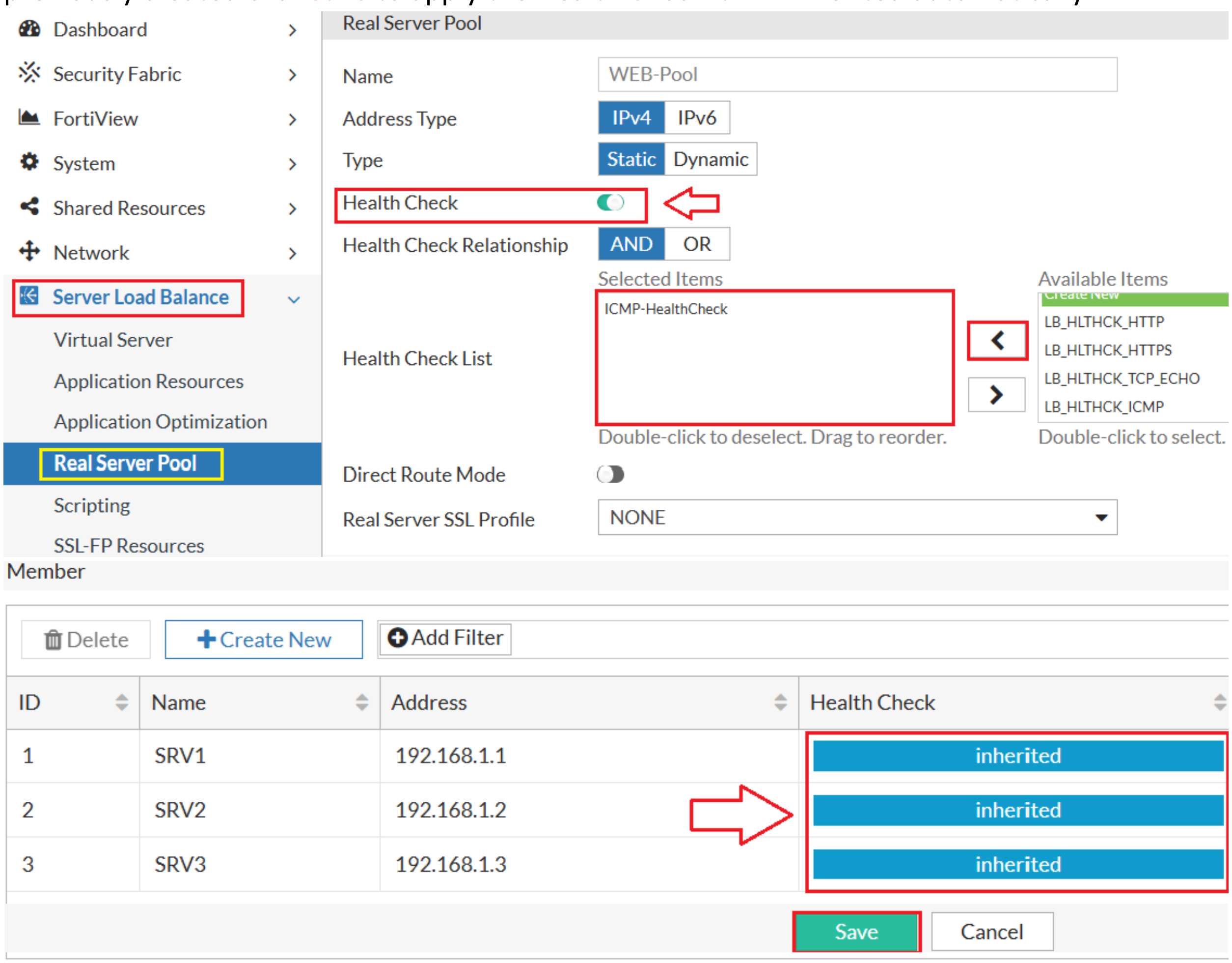
Type Name in this case ICMP-HealthCheck and choose Type ICMP click Save.



Next navigate to Server Load Balance> Real Sever Pool > Real Server Pool tab double click on WEB-Pool to open the setting.



Toggle to enable the Health Check on right side Double-click to choose ICMP-HealthCheck previously created click Save to apply the Health Check it will inherited automatically.



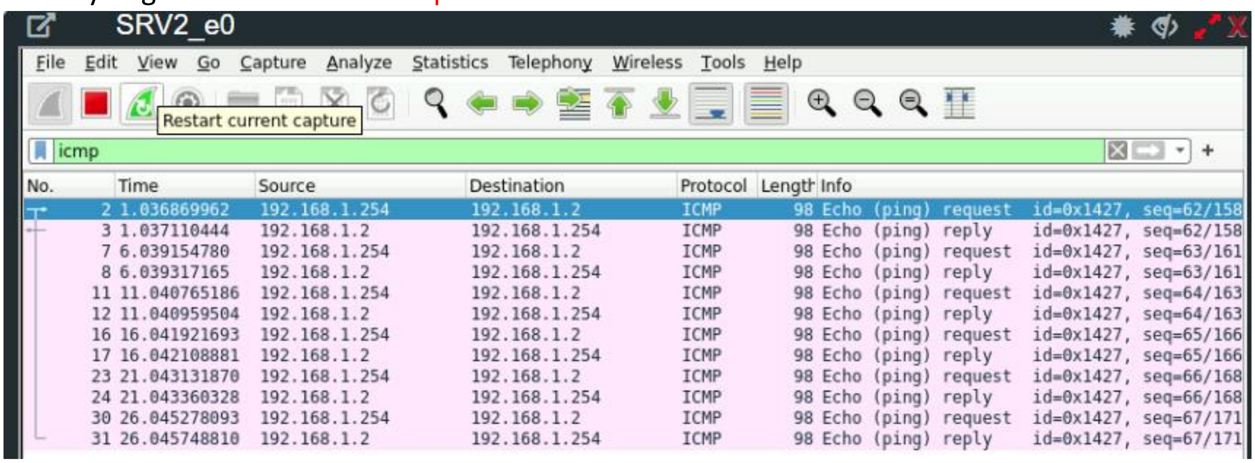
 Created by Ahmad Ali E-Mail: ahmadalimsc@gmail.com , WhatsApp: 00966564303717

Verification:

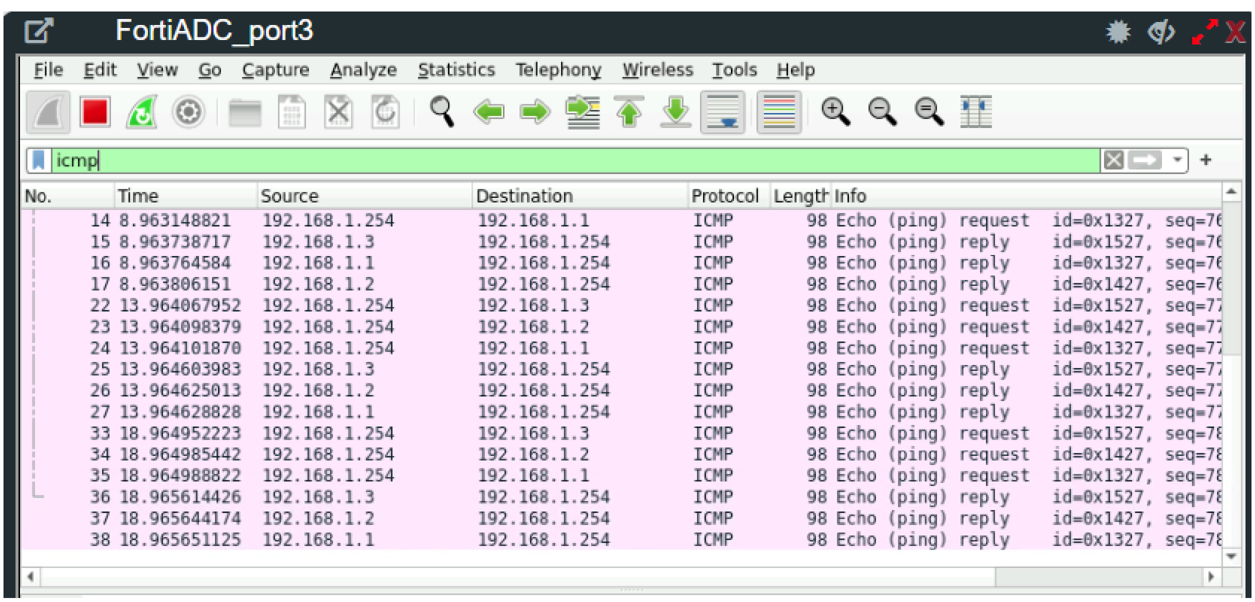
Go to SRV1 login with username & password root/root type tcpdump icmp.

```
SRV1
root@SRV1:~# tcpdump icmp
tcpdump: verbose output suppressed, use -v or -vv for full protocol
listening on ens3, link-type EN10MB (Ethernet), capture size 262144
17:53:53.388328 IP 192.168.1.254 > 192.168.1.1: ICMP echo request,
17:53:53.388349 IP 192.168.1.1 > 192.168.1.254: ICMP echo reply, id
17:53:58.390331 IP 192.168.1.254 > 192.168.1.1: ICMP echo request,
17:53:58.390350 IP 192.168.1.1 > 192.168.1.254: ICMP echo reply, id
17:54:03.391584 IP 192.168.1.254 > 192.168.1.1: ICMP echo request,
17:54:03.391600 IP 192.168.1.1 > 192.168.1.254: ICMP echo reply, id
^C
```

Alternatively, you can use the built-in Wireshark capture tool in EVE-NG to monitor ICMP traffic directly. Right click one SRV2 >Capture choose interface e0



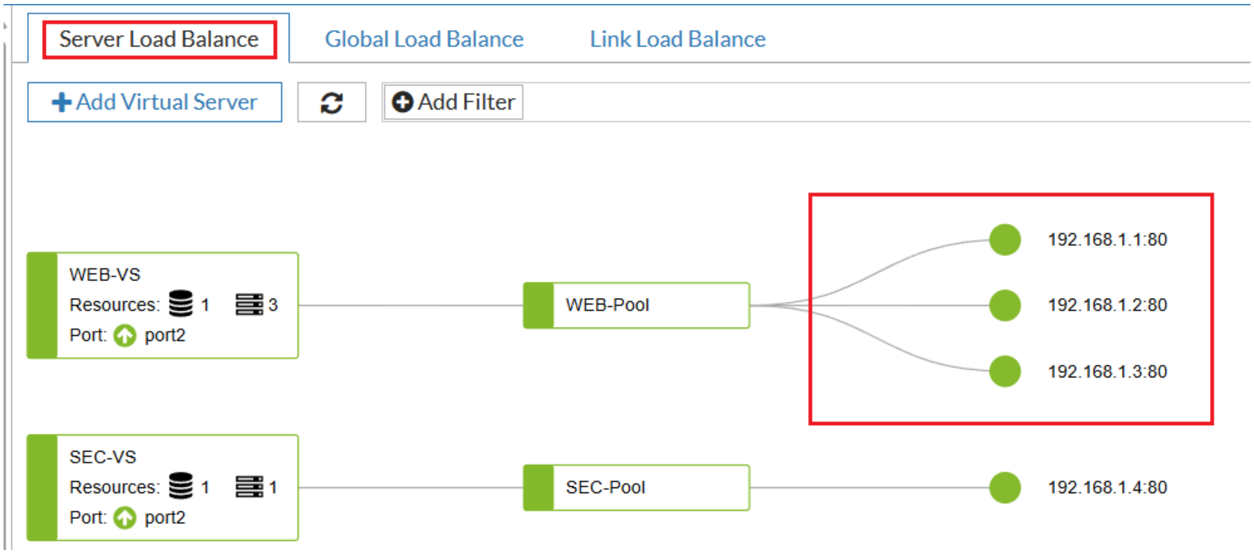
No.	Time	Source	Destination	Protocol	Length	Info
2	1.036869962	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=62/158
3	1.037110444	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=62/158
7	6.039154780	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=63/161
8	6.039317165	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=63/161
11	11.040765186	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=64/163
12	11.040959504	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=64/163
16	16.041921693	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=65/166
17	16.042108881	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=65/166
23	21.043131870	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=66/168
24	21.043360328	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=66/168
30	26.045278093	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=67/171
31	26.045748810	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=67/171



No.	Time	Source	Destination	Protocol	Length	Info
14	8.963148821	192.168.1.254	192.168.1.1	ICMP	98	Echo (ping) request id=0x1327, seq=76
15	8.963738717	192.168.1.3	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1527, seq=76
16	8.963764584	192.168.1.1	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1327, seq=76
17	8.963806151	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=76
22	13.964067952	192.168.1.254	192.168.1.3	ICMP	98	Echo (ping) request id=0x1527, seq=77
23	13.964098379	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=77
24	13.964101870	192.168.1.254	192.168.1.1	ICMP	98	Echo (ping) request id=0x1327, seq=77
25	13.964603983	192.168.1.3	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1527, seq=77
26	13.964625013	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=77
27	13.964628828	192.168.1.1	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1327, seq=77
33	18.964952223	192.168.1.254	192.168.1.3	ICMP	98	Echo (ping) request id=0x1527, seq=78
34	18.964985442	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=78
35	18.964988822	192.168.1.254	192.168.1.1	ICMP	98	Echo (ping) request id=0x1327, seq=78
36	18.965614426	192.168.1.3	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1527, seq=78
37	18.965644174	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=78
38	18.965651125	192.168.1.1	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1327, seq=78

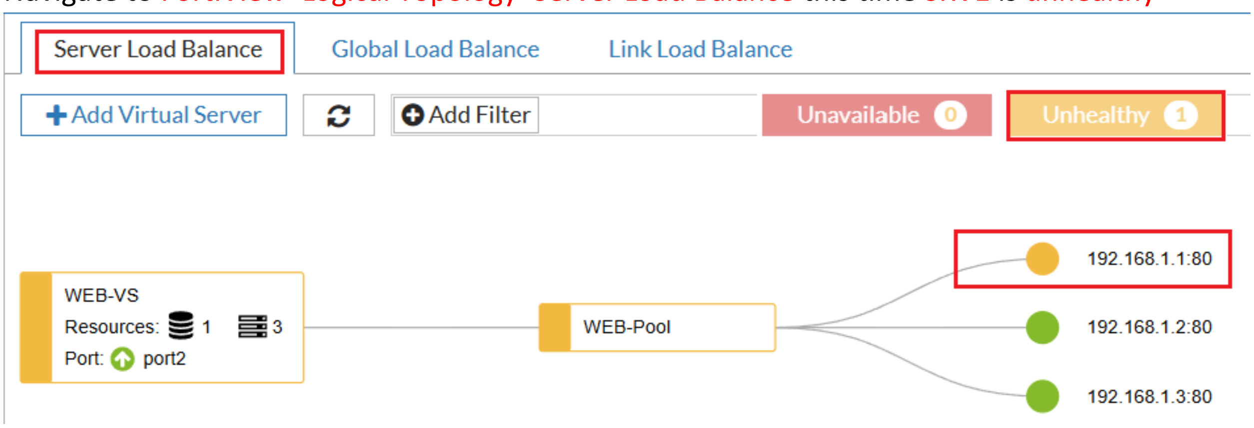
 Created by Ahmad Ali E-Mail: ahmadalimsc@gmail.com , WhatsApp: 00966564303717

Navigate to FortiView>Logical Topology>Server Load Balance all the servers are up green and healthy.

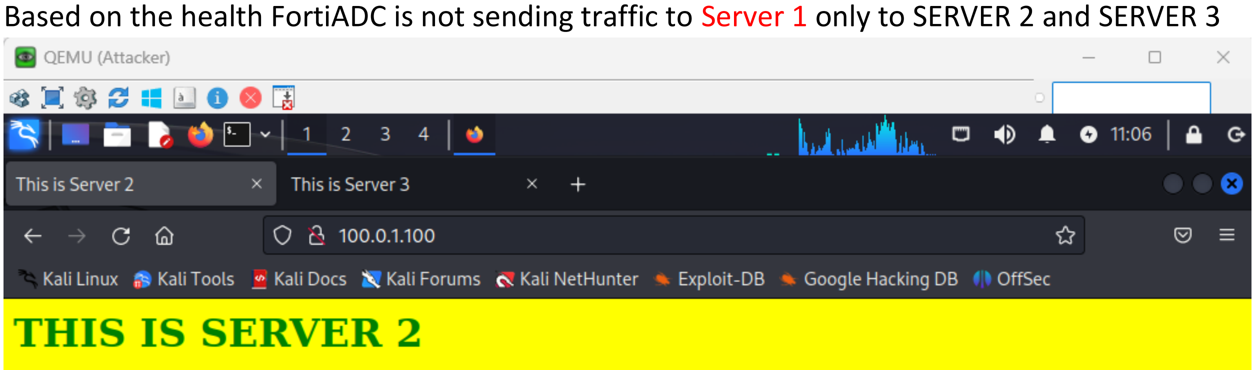


To test the ICMP-Based Health check let's stop one of the Server in this case SRV1.
root@SRV1:~# shutdown -h now

Navigate to FortiView>Logical Topology>Server Load Balance this time SRV1 is unhealthy

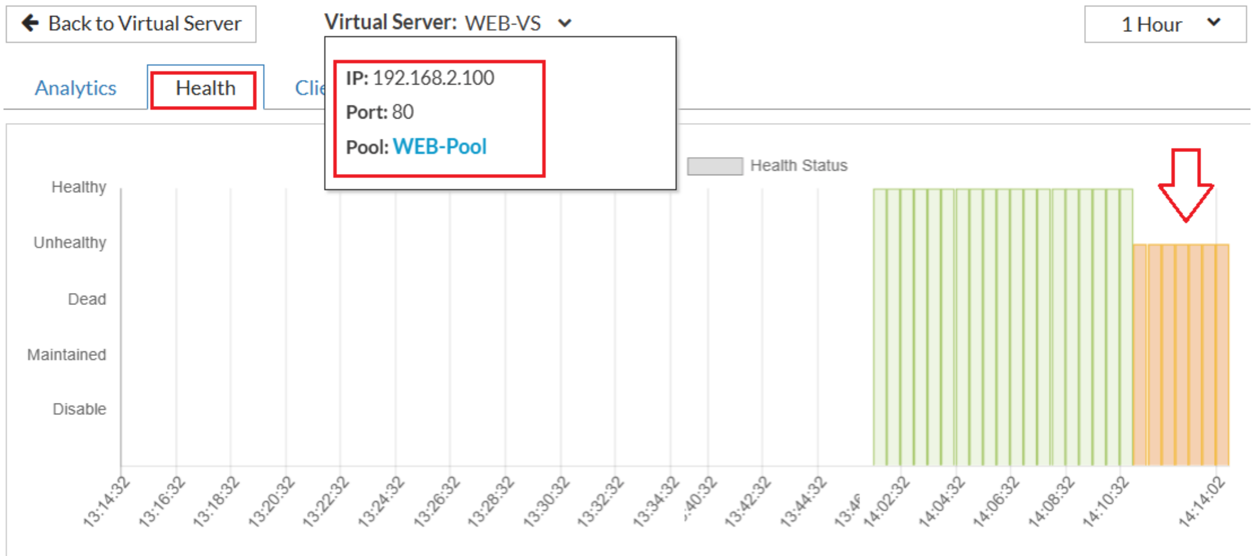


Based on the health FortiADC is not sending traffic to Server 1 only to SERVER 2 and SERVER 3

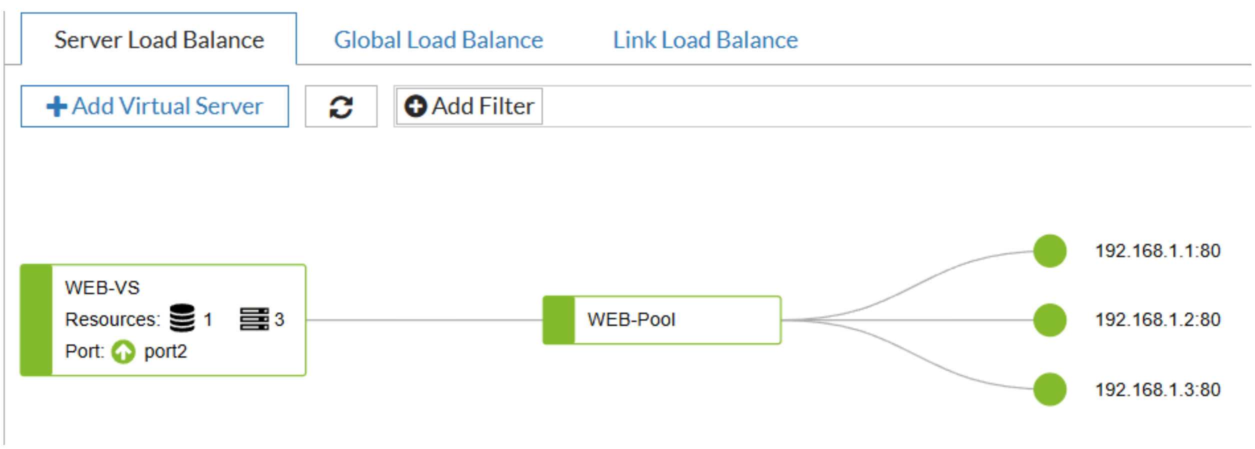


 Created by Ahmad Ali E-Mail: ahmadalimsc@gmail.com , WhatsApp: 00966564303717

Navigate to FortiView>Virtual Server click on the name of Virtual Server in this case WEB-VS after that click on Health tab to check Health Status the organ color show is unhealthy



Let's start SERV1, but this time, stop only the Web Services running on it. This approach allows us to simulate a scenario where the server itself is operational and can respond to basic network probes, such as ICMP (ping), but the application or web service is unavailable.
root@SRV1:~# service apache2 stop



In this case ICMP-based health check were used, FortiADC not able to detect the failure on Servers. While the web service on servers are down, the server is still capable of responding to ICMP probe packets. Consequently, FortiADC would interpret the server as healthy, as it relies on the ICMP replies for status verification, missing the underlying application-level issue. This highlights the importance of using health checks that are specific to the application's functionality for accurate failure detection.